

THIRUVENKADAPRASATH.S

192312016

9	Compare Linear and Polynomial Regression using Python
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CSV FILE

Position_Level,Salary

1,45000

2,50000

3,60000

4,80000

5,110000

6,150000

7,200000

8,300000

9,500000

10,1000000

PROGRAM

```
import numpy as np
```

```
import pandas as pd
```

```
class LinearRegressionScratch:
```

```
    def __init__(self):
```

```
        self.weights = None
```

```
    def fit(self, X, y):
```

```
        X_b = np.c_[np.ones((len(X), 1)), X]
```

```
self.weights = np.linalg.inv(X_b.T.dot(X_b)).dot(X_b.T).dot(y)
```

```
def predict(self, X):
```

```
    X_b = np.c_[np.ones((len(X), 1)), X]
```

```
    return X_b.dot(self.weights)
```

```
def create_polynomial_features(X, degree):
```

```
    X_poly = X.copy()
```

```
    for d in range(2, degree + 1):
```

```
        X_poly = np.c_[X_poly, X**d]
```

```
    return X_poly
```

```
df = pd.read_csv("regression_comparison_data.csv")
```

```
X = df[["Position_Level"]].values
```

```
y = df["Salary"].values
```

```
linear_model = LinearRegressionScratch()
```

```
linear_model.fit(X, y)
```

```
y_pred_linear = linear_model.predict(X)
```

```
degree = 4
```

```
X_poly = create_polynomial_features(X, degree)
```

```
poly_model = LinearRegressionScratch()
```

```
poly_model.fit(X_poly, y)
```

```
y_pred_poly = poly_model.predict(X_poly)
```

```

r2_linear = 1 - (
    np.sum((y - y_pred_linear) ** 2) / np.sum((y - np.mean(y)) ** 2)
)
r2_poly = 1 - (np.sum((y - y_pred_poly) ** 2) / np.sum((y - np.mean(y)) ** 2))

mse_linear = np.mean((y - y_pred_linear) ** 2)
mse_poly = np.mean((y - y_pred_poly) ** 2)

print(f"--- Linear Regression Performance ---")
print(f"Mean Squared Error: {mse_linear:.2f}")
print(f"R2 Score: {r2_linear:.4f}\n")

print(f"--- Polynomial Regression (Degree {degree}) Performance ---")
print(f"Mean Squared Error: {mse_poly:.2f}")
print(f"R2 Score: {r2_poly:.4f}\n")

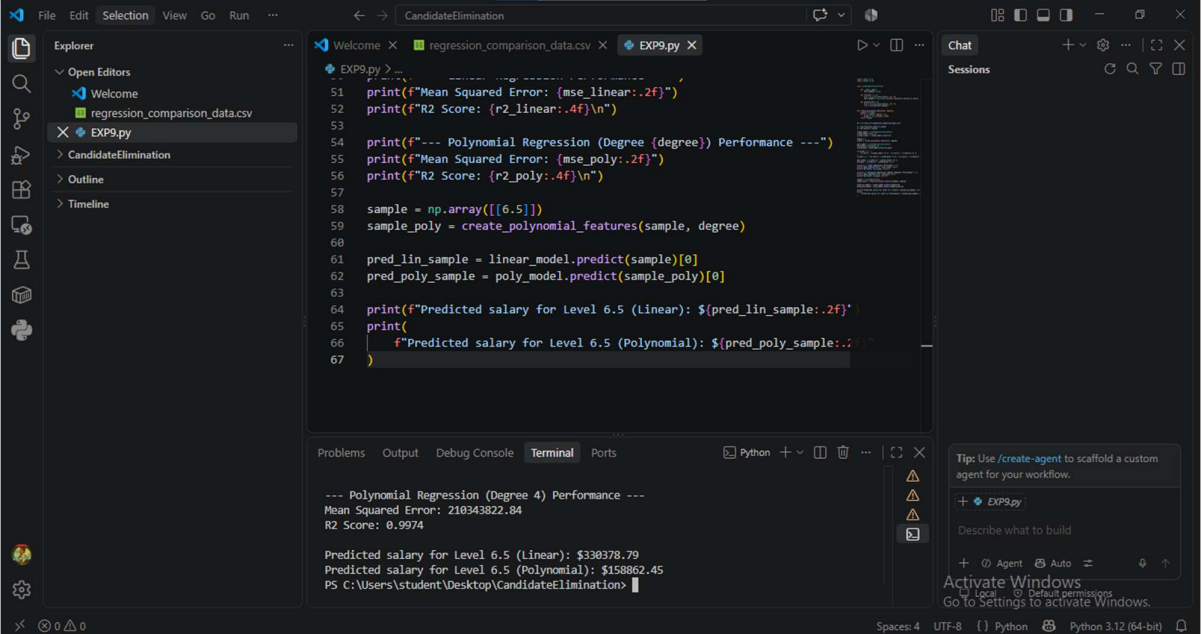
sample = np.array([[6.5]])
sample_poly = create_polynomial_features(sample, degree)

pred_lin_sample = linear_model.predict(sample)[0]
pred_poly_sample = poly_model.predict(sample_poly)[0]

print(f"Predicted salary for Level 6.5 (Linear): ${pred_lin_sample:.2f}")
print(
    f"Predicted salary for Level 6.5 (Polynomial): ${pred_poly_sample:.2f}"
)

```

OUTPUT



The screenshot shows a Visual Studio Code editor window with a Python script named `EXP9.py` open. The script performs polynomial regression on a dataset, comparing linear and polynomial models. The output is displayed in the terminal window at the bottom.

```
51 print(f"Mean Squared Error: {mse_linear:.2f}")
52 print(f"R2 Score: {r2_linear:.4f}\n")
53
54 print(f"--- Polynomial Regression (Degree {degree}) Performance ---")
55 print(f"Mean Squared Error: {mse_poly:.2f}")
56 print(f"R2 Score: {r2_poly:.4f}\n")
57
58 sample = np.array([[6.5]])
59 sample_poly = create_polynomial_features(sample, degree)
60
61 pred_lin_sample = linear_model.predict(sample)[0]
62 pred_poly_sample = poly_model.predict(sample_poly)[0]
63
64 print(f"Predicted salary for Level 6.5 (Linear): ${pred_lin_sample:.2f}")
65 print(
66     f"Predicted salary for Level 6.5 (Polynomial): ${pred_poly_sample:.2f}"
67 )
```

The terminal output shows the results of the polynomial regression (Degree 4) performance:

```
--- Polynomial Regression (Degree 4) Performance ---
Mean Squared Error: 210343822.84
R2 Score: 0.9974

Predicted salary for Level 6.5 (Linear): $330378.79
Predicted salary for Level 6.5 (Polynomial): $158862.45
PS C:\Users\student\Desktop\CandidateElimination>
```